THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

REVIEW & CORRECTION REPORT

Existing Class 10 Mathematics worksheets (Ch 1–4, 5–7, 8–11, 12–14) checked against the **RBSE Model Paper 2026–27** and the rationalised NCERT syllabus

Summary: the structure is right (40 questions, 20/16/24/20 marks, internal choice in Q35–40, a graph question at Q29), and most answers are correct. But **12 answers or figures in the keys and blueprint are wrong** (7 in the Ch 12–14 paper alone). Two questions come from sections dropped from the syllabus, several questions repeat within the same paper, and some OR options have no answers in the key. All of these must be fixed before the papers are printed.

Ch 1–4 (Real Numbers, Polynomials, Linear Equations, Quadratic Equations) — 11 items

Q.	Severity	Problem found	Correction
4	Syllabus	Decimal expansion of $\frac{17}{8}$ (terminating / non-terminating). This section was dropped from the rationalised NCERT book that RBSE follows.	Replace with an FTA / irrationality item.
Blue-print	Wrong key	Chapter-wise weightage shows 12 + 14 + 16 + 18 marks, which adds to 60, not 80. The 'Total' row is also printed white-on-white (invisible).	Actual marks in the paper: Ch1 = 10, Ch2 = 18, Ch3 = 25, Ch4 = 27 (total 80).
17 & 24	Duplicate	Same question twice: "Both roots of $3x^2 - 12x + 12 = 0$ are not equal".	Change one of them.
8 & 23	Duplicate	Same pair $3x + 2y = 5,\ 2x - 3y = 7$ in both.	Change Q23.
16 & 34	Duplicate	Same equation $2x^2 - 4x + 3 = 0$.	Change Q34.
11 & 36	Duplicate	Same pair $x + 2y - 4 = 0,\ 2x + 4y - 12 = 0$.	Change Q36.
36-OR & 38-OR	Duplicate	Both use $9x - 4y = 2000,\ 7x - 3y = 2000$.	Change one.
2 & 35	Duplicate	Same polynomial $x^2 - 5x + 6$.	Change Q35.
35, 36, 39 (OR)	Missing	The answer key has no solutions for the OR options.	35-OR: $x^2 + 3x + 2$; 36-OR: $x = 2000,\ y = 4000$; 39-OR: zeroes $\frac{1}{2}, \frac{1}{2}$.
4, 19	Layout	Exponents ($2^m \times 5^n$, x^2 in the options) float above the line instead of sitting as superscripts.	Re-typeset.
All	Layout	Page breaks leave half-empty pages (page 2 holds only Q20), so the paper runs to 13 pages.	—

Ch 5–7 (Arithmetic Progressions, Triangles, Coordinate Geometry) — 8 items

Q.	Severity	Problem found	Correction
9	Wrong key	Key says (b) the points $(1,5), (2,3), (-2,-11)$ are **collinear**. They are not ($\sqrt{5} + \sqrt{212} \neq \sqrt{265}$), and the paper's own Q34 key says "not collinear".	Correct answer: **(a) vertices of a triangle**.
14	Syllabus	Ratio of areas of similar triangles: the 'areas of similar triangles' section was dropped from the rationalised syllabus.	Replace, e.g. with a perimeter ratio or a BPT item.
37-OR	Wrong key	Asks to "verify that the total interest in 30 years equals the sum of the AP" and the key gives ₹37,200. The terms 80, 160, 240, … are already the *cumulative* interest at the end of each year, so the total interest after 30 years is ₹2400. Adding the terms means nothing.	Delete the 'verify' part; the answer is ₹2400.
17 & 30	Duplicate	Same ladder (rungs 25 cm apart, 45 cm to 25 cm) used twice.	Change Q17.

16 & 27	**Duplicate**	Both use P(2, −3), Q(10, y), distance 10.	Change Q27.
9 & 34	**Duplicate**	Same three points.	Change one.
35-OR, 36-OR	**Missing**	No key for the OR options.	35-OR: $n = 16,\ d = \frac{8}{3}$; 36-OR: $x = \pm 4$, $QR = \sqrt{41}$, $PR = \sqrt{82}$ (for $x = 4$) or $9\sqrt{2}$ (for $x = -4$).
38 (iii)	**Wording**	Key justifies △CMB ~ △RNQ by 'SSS' without showing it.	Use SAS: $\angle B = \angle Q$ and $\frac{BC}{QR} = \frac{BM}{QN}$.

Ch 8–11 (Trigonometry, Applications, Circles, Areas Related to Circles) — 6 items

Q.	Severity	Problem found	Correction
18	**Wrong key**	"Angle 120°, area of the **major** sector 462 cm², find r". Key: 21 cm. A 240° major sector of 462 cm² gives $r^2 = 220.5$, so $r \approx 14.85$ cm, which is not among the options.	Change 'major sector' to 'sector' (a 120° sector of 462 cm² gives $r = 21$ cm).
40-OR	**Wrong key**	Wiper area: key gives 627.08 per wiper and 1254.16 in total.	$\frac{115}{360} \times \frac{22}{7} \times 625 = 627.48$ cm²; total **1254.96 cm²** (still about 4 sweeps).
34	**Wording**	Key writes 'Area △OAB = $\frac{\sqrt{3}}{4} \times 21^2$', the equilateral-triangle formula, for a 120° triangle. The value 190.73 is right only by coincidence.	Use $\frac{1}{2}r^2 \sin 120^\circ$ or drop a perpendicular from O.
35-OR, 36-OR	**Missing**	No key for the OR options.	35-OR: building $= \frac{50}{3} \approx 16.67$ m; 36-OR: designs $= 464.8$ cm², cost **₹162.68**.
8	**Wording**	Option (a) reads "Always true (alternate angles)", which gives the answer away.	Remove the bracket.
Fig. 31	**Layout**	Labels P and A overlap; C is hidden at the foot of the tower.	Redraw.

Ch 12–14 (Surface Areas & Volumes, Statistics, Probability) — 10 items

Q.	Severity	Problem found	Correction
2	**Wrong key**	Key marks (b) 77.6 cm³ and then contradicts itself in a note.	$\frac{1}{3} \times \frac{22}{7} \times 2.1^2 \times 4.2 = 19.4$ cm³ ⇒ **(d)**.
3	**Wrong key**	Two cylinders stacked: key (a) $4\pi rh + 4\pi r^2$.	New solid is a cylinder of height $2h$: $4\pi rh + 2\pi r^2$ ⇒ **(b)**.
4	**Wrong key**	Capsule surface area: key (a) 110 mm².	$2\pi(2.5)(9) + 4\pi(2.5)^2 = 70\pi = 220$ mm² ⇒ **(b)** (NCERT answer).
17	**Wrong key**	Key says odd and even outcomes of a die are "not equally likely". They are (3 faces each), so $P(\text{odd}) = \frac{1}{2}$ is right. The question doesn't work as written.	Replace with NCERT's "11 possible sums of two dice" argument.
24	**Wrong key**	Median 28.5, find x: key gives $x = 5$.	$\frac{(45+x)/2-(5+x)}{20} \times 10 = 8.5 \Rightarrow x = 1$ (check: $n = 46$, $20 + \frac{23-6}{20} \times 10 = 28.5$).
35	**Wrong key**	Iron pole: key gives $V_1 = 99532.8$ and a total of 115590.4 (addition error); mass 924.7 kg.	$V_1 = 99475.2$, $V_2 = 12057.6$, total 111532.8 cm³ ⇒ mass ≈ **892.26 kg** (π = 3.14).
38-OR	**Wrong key**	Mean weight by step deviation: key 56.83 kg.	$\Sigma fu = -6 - 6 - 8 + 0 + 6 + 6 + 6 = -2$ ⇒ mean $= 57.5 - \frac{1}{3} =$ **57.17 kg** (median 56.67 is right).
33 & 38	**Duplicate**	Identical electricity data (68 consumers) in both.	Change Q33.
34 & 40-OR	**Duplicate**	Identical '90 discs' question.	Change one.
1–5, 12, 19	**Layout**	Superscripts in cm², cm³, r^2 and the subscripts f_0, f_1, f_2 in the mode formula are detached, which makes Q5 unreadable.	Re-typeset.